

Imbalanced Data

⇒ What is imbalanced dataset? In such dataset one type of data is too much than the other.

⇒ Problems with imbalanced data?

- ① Bias towards majority class
- ② Certain metrics are not reliable, like accuracy for classification problems.

⇒ How to solve the issue of imbalanced dataset?

- ① Undersampling
- ② Oversampling
- ③ SMOTE
- ④ Ensemble methods - Balanced Random Forest
- ⑤ Cost sensitive Learning.

Optuna

⇒ Why Optuna? Bez Grid Search CV is costly and random search CV can miss important things. Hence, we need Optuna. It uses Bayesian Search.

This search actually establish a relation b/w max-depth and n -estimator. i.e.,

$$\text{accuracy} = f(\text{max-depth}, n\text{-estimator})$$

and find maximo for it.